Nudging in Public Policy: A Review of Empirical Studies

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Introduction

Behavioral economics is a frontier area of economic research. Challenging the original assumptions of rationality, behavioral economics expands and generalizes psychological bias into the context of traditional financial analysis. Instead of subverting the long-lasting foundation and the whole economic framework, behavioral economics is an excellent supplement for traditional economics, bringing theory into real-world decision-making.

In recent decades, behavioral economics has also induced robust insights in public policy, providing useful policy tools to influence individual decisions. "Nudges" are one of these tools, introduced by Thaler & Sunstein (2009). They proposed the idea of "Libertarian paternalism" as "tries to influence choices in a way that will make choosers better off, as judged by themselves" (Thaler & Sunstein, 2003). A "nudge" is implemented by policy-makers as choice architecture, by giving individuals the liberty to change their options without offering financial incentives. Although nudges are frequently cost-effective, doable, and provocative as a policy tool, they also generate important questions concerning unintended consequences and long-term welfare effects: Under what conditions do nudges yield better outcomes than conventional instruments such as taxes and subsidies? Are all nudges good for society? Although nudging has been extensively studied in theory, empirical evidence is sparse due to the difficulty and high cost of maintaining an experiment.

In this paper, I attempt to expand the idea of nudging from behavioral economic theory to policy evidence by integrating recent empirical research to explore how nudging compares with older-fashioned economic interventions and where and when it might be tried or fail to be tried. In the first section, I will discuss the role of "Libertarian Paternalism" and its relationship with

"Nudging". In the following two sections, I will take two empirical papers as examples to discuss when and how nudging, which is implemented to increase social welfare, turned out to backfire. Then, I will extend the discussion about "Nudging" in Behavioral economics. Eventually, I will end the conversation about nudging by discussing the limitations and policy implications

Libertarian Paternalism and Nudge

Thaler & Sunstein (2003) described “Libertarian Paternalism” as a policy approach “that preserves freedom of choice but that authorizes both private and public institutions to steer people in directions that will promote their welfare.” They proposed this idea to bridge two seemingly contradictory concepts, “libertarian” and “paternalism”. Seen from the literal expression, the word libertarian “ emphasized the freedom of choice, while “paternalism“ stressed the idea of taking control of someone and restricting their freedom in order to improve their welfare. However, Thaler & Sunstein contended that these two concepts are not mutually exclusive. By thoroughly and intentionally devising the choice architecture, the policymaker can direct people's thoughts and improve their welfare without manipulating the pricing mechanism. This concept challenged traditional economics in the sense of human rationality and the role of choice architecture.

Fundamentally, libertarian paternalism departs from traditional economics as it rejects the assumptions of rationality. Traditional economic theory assumes people maximize their utility with full information and unlimited cognitive abilities, and their revealed preferences satisfy completeness and consistency (Binmore, 2008). Thaler and Sunstein contended that this assumption has been questioned, and a wealth of research from psychology and behavioral economics has demonstrated that individuals systematically deviate from perfect rationality. People utilize heuristics that can lead to systematic errors, exhibit preference reversals, and make different choices depending on the wording of the problem. Furthermore, people often struggle with self-control problems, particularly in intertemporal choices. The other key difference arises from libertarian paternalists asserting that some form of paternalism is often inevitable and that any set-up of choices will influence choices. In contrast, traditional economists tend to assume the setup of choices has an insignificant effect on how people make decisions and believe people will constantly rationally choose the best option. This is the concept of "choice architecture".

A policy is paternalistic if it aims to influence choices to make the chooser better off, when the policy makers acknowledge and intend to mend the gap between revealed preference and welfare. In the libertarian paternalist's toolbox, Thaler and Sunstein suggest approaches such as:

(1) Mimicking informed choices: Selecting the approach that the majority would choose if they made an explicit, informed decision. (2) Forcing active choices: Requiring individuals to choose explicitly, rather than defaulting them into an option. (3) Minimizing opt-outs: Minimizing the number of opt-outs if staying in is more favored.

A nudge intervention is a salient application of libertarian paternalism. List et al (2023) coined "Nudge" as "Non-price interventions motivated by insights from psychology." In other words, it accepts that some form of choice-shaping is inevitable, yet insists on maintaining individuals' ability to opt out of any arrangement. The following properties characterize a nudge (see also Rebonato 2012, p. 32; Till Grune-Yanoff1 & Ralph Hertwig, 2016):

1. A nudge is intended by the policy maker (choice architect) to steer the chooser's behavior away from the behavior implied by the cognitive shortcoming and toward her ultimate goal or preference (e.g., healthier food choices).

2. A nudge seeks to realize this influence by exploiting empirically documented cognitive shortcomings in human deliberation and choice, without changing the financial incentives (disincentives).

3. A nudge does not affect those features over which people have explicit preferences (e.g., money, convenience, taste, status, etc.), but rather those features that people would typically claim not to care about (e.g., position in a list, default, framing).

4. The behavior change brought about by the nudge should be easily reversible, allowing the chooser to act otherwise.

Ralph Hertwig (2016) provides examples applying nudge in public policy, including:

1. Setting Defaults: Determining the option an individual receives if they do not make an active choice. Take a 401 (k) as an example. Automatic enrollment was made in retirement savings. Economists attributed its effectiveness to people's inertia, status quo bias, heuristics, and loss aversion.

2. Changing Time Horizons: The "Save More Tomorrow" program increases retirement contributions by asking for consumption between the near future and later, instead of a trade-off between consumption now and later. This took advantage of time-inconsistent preferences (overvaluing the present) and inertia.

3. Framing is the process of presenting information in a way that influences choice. This will impact people's decisions by changing how they perceive the question without changing the content because people are often "mindless, passive decision makers."

To summarize, nudges take advantage of psychological biases and work on fast, automatic, and often unconscious "Type 1" or "System 1" cognitive processes. The "libertarian" aspect emphasizes that people remain free to choose, while the "paternalistic" aspect is the intention to make choosers better off, judged by their criteria. (Thaler & Sunstein, 2008; Banerjee & John, 2024)

Research Study: Judging nudging: Understanding the welfare effects of nudges versus taxes

List et al (2023) compared "nudges" and traditional policies with financial incentives like taxes and subsidies by pioneering a framework as general criteria. Their work is valuable as they quantify the indistinguishable and ambiguous economic welfare and critically analyse whether the nudge is a better policy.

Internalities and Externalities

They endeavored to express consumer behavior and social welfare by adding two distortions, market internalities and externalities. In the classic terminology of welfare economics, externalities exist when an individual's action imposes a cost or a benefit on others who do not belong to any party in the transaction, and this spillover isn't considered in the market price. A classic example would be the water pollution from a manufacturing factory or the smell of waffle trucks. The personal consumer doesn't have to pay for the environmental harm problems experienced by others. Vaccination also has its externality. By getting vaccinated, you protect yourself and everybody else from disease risk. Since people think only about their private costs and benefits, goods with negative externalities get consumed excessively, and those with positive externalities are under-consumed from the perspective of society.

On the other hand, internalities are a core concept of behavioral economics, and they occur when people harm themselves because of "systematic mistakes" in choices. These mistakes are not random but the result of predictable psychological biases. They make decisions that are not in their long-term interests. For example, smokers might keep on underestimating the extent to which their habit will affect their future health because of overconfidence bias or self-control problems (present bias). Households may overconsume energy because they are inattentive to the costs of appliances or miscalculate potential savings from conservation. As such, internalities cause people to "choose" paths of action that make them unhappier in the long run. Hence, internalities indicate a case for "paternalistic" interventions that can help them do better for themselves.

Externalities and internalities distort economic welfare by creating a wedge between what people choose to maximize their utility and what would be most beneficial for society, or even for the people themselves in the long run. Such a misfit means a waste of economic efficiency, resulting in a waste of overall economic efficiency. Therefore, policymakers endeavored to address this misfit, which is called “deadweight loss” in economics. In the best possible world, the policy would alleviate both distortions. Yet taxes merely correct externalities by incorporating private costs into social costs, while nudges try to realign individuals' perceptions of the world to reality, therefore correcting internalities.

The Nudging Mechanism

The paper explicitly crystallized welfare and modeled it in economic models by considering three ways nudges can influence welfare as follows:

1. **The Benchmark View: Nudges Correct Biases ("De-biasing")**: This is the most common statement for nudging, and is treated as the primary "benchmark" model in this paper. In this model, nudges reduce or eliminate the behavioral biases that distort judgment. For example, reminders make people more attentive, clear information makes people correct their misconceptions, and simplified choices reduce cognitive load. The

result of behavioral change is assumed to represent an improvement toward choices that align with the individual's underlying preferences and long-term goals. Under this view, nudges directly improve individual welfare by reducing the harm caused by internalities.

2. **The Skeptical View: Nudges Deceive**: Sometimes, instead of correcting a bias, a nudge itself creates a new bias, or pushes people away from what would have been the best choice. For instance, a warning may cause smokers to overestimate the health risks beyond what is true, causing them to quit smoking. Such a “deceptive” nudge might even lower smoking rates. It enhances economic welfare because it reduces externalities and raises social welfare. But as for their personal preference, they may gain more welfare from smoking. In the meantime, welfare decreases as it biases a subject's choice away from their optimal preference, and net welfare becomes indeterminate. In that sense, it may be that a nudge backfires by creating some new set of psychological costs or benefits not directly tied to real-world results.

3. **The Affective View: Nudges Impose Moral Costs or Benefits**: Nudges place moral costs or benefits on options rather than changing beliefs or correcting biases. One mechanism through which a nudge might operate is to prompt emotions such as guilt, shame, pride, or satisfaction. For instance, a graphic warning label may make smokers feel unpleasant or shameful. A report indicating that you use less energy than your neighbors could make you feel good. In this model, the nudge affects utility apart from the consumption change.

The Theoretical trade-off: Nudge or Taxes?

The most significant contribution in this paper is to identify theoretical trade-offs that determine whether nudges or taxes are likely more efficient in different markets. "While both nudges and taxes can correct the average bias of marginal consumers, only nudges can reduce variance in bias, and only taxes can internalize the externality." (List et al., 2023) Taxes and nudges address different sources of market inefficiency, so they may perform differently and have pros and cons when deciding which instruments to use.

The first trade-off they identify is that taxes excel at correcting the average. The paper pointed out that taxes are more efficient if there is an enormous average discrepancy between private and social costs/benefits. The reason is that although nudge can affect the total level of externality through changing the consumption, the difference between social and private benefits remains unchanged among marginal consumers, who are sensitive to price change. However, a single tax rate can deal with the problem relatively well. For example, nudging in the energy market reduces energy consumption but has little effect on internalizing the external damage of carbon emissions.

On the other hand, simply imposing a carbon tax can internalize the marginal environmental damage and solve the problem. The other trade-off is that nudge corrects heterogeneity. A uniform tax treats every consumer the same, but behavioral biases can vary significantly across individuals in the real world. De-biasing nudge, the benchmark model in the paper, has a competitive advantage. They help individuals overcome their specific bias by providing more information, thus reducing individual-specific bias.

The paper crystallizes this trade-off between nudges and taxes into a concept called the "targeting ratio." Intuitively, this ratio compares the size of the problem caused by the variation in behavioral biases (specifically, the standard deviation of the bias, weighted by consumer price sensitivity) to the size of the problem caused by the average externality. Suppose the targeting ratio is greater than one. In that case, the inefficiency from bias heterogeneity is the dominant issue, and then nudges that effectively reduce heterogeneity are predicted to enhance welfare more than taxes. In contrast, if the targeting ratio is less than one, the average externality is a bigger problem. In this case, taxes directly targeting this average externality are predicted to be more efficient. This ratio conceptualizes the trade-off and makes it a decisive criterion to consider when and why one policy approach might be preferred. Despite its usefulness, the actual value of this ratio is an empirical question that depends on the specific market.

Empirical evidence

This paper is also insightful because it provides data-based evidence to estimate the target ratio and overall welfare effects. The researchers gathered information from 304 published studies that estimate either the impact of a particular nudge on a behavior or the effect of a price change on a behavior. These studies were carefully screened and coded into three markets of interest: cigarettes, influenza vaccines, and household electricity. Then, they calculated the average effect of nudges, the average price sensitivity in each market, and the variation in these effects across studies. The estimated variation in nudge can be interpreted as a measure of heterogeneity in behavioral bias. They combined these estimates with external estimates of the average externality in each market, specifically, the health costs of smoking, the herd immunity benefits of vaccines, and the social cost of carbon for electricity. These estimates allowed them to incorporate empirical values into their theoretical welfare analysis.

In the cigarette market, nudges appear to be superior. The data suggested that while smoking has negative externalities, the variation in behavioral biases among smokers is an even larger source of inefficiency. People smoke too much for many different reasons. They differ in levels of addiction, self-control, and risk perception. A uniform cigarette tax helps reduce overall smoking, but doesn't target these diverse underlying problems as effectively as nudges potentially can. Instead, nudges consistently showed higher estimated welfare gains than the optimal tax. The targeting ratio was estimated at 1.9, significantly above one, supporting the idea that heterogeneity is the dominant issue. In addition, combining a nudge with a tax offered minimal additional welfare gain. This suggests the nudge was already addressing the primary source of inefficiency.

In the influenza vaccinations market, however, subsidies are likely superior. Getting a flu shot generates substantial positive externalities through herd immunity. People may under-vaccinate due to biases like procrastination or underestimating personal risk. Yet, the optimal subsidy yielded higher welfare gains than the average nudge under most assumptions. The targeting ratio was estimated at 0.46, less than one. Despite the significance, the researchers stated that the results might be harmed by a limited number of price elasticity studies found for vaccines.

As for household electricity, taxes are vastly superior. Electricity generation, particularly from fossil fuels, carries a huge negative externality via climate change. While households may exhibit biases leading to some overconsumption, this internal problem is minimal compared to the externality problem. The estimated welfare gains from an optimal electricity tax were enormous and decreased the gains from nudges by a factor of 7 to 9. This finding was extremely robust across all tested correlation and nudge effectiveness assumptions. The targeting ratio was very low (0.069), confirming the dominance of the externality. Adding nudges to the optimal tax provided almost no additional benefit.

This paper also investigates the sensitivity of the benchmark findings to alternative models of how nudges affect welfare. From these specifications, the conclusion remained unchanged.

From the "Deceptive Nudges" point of view, nudges that might mislead consumers, either by exacerbating existing biases or reversing their signs. If nudges exacerbate biases, they strictly reduce welfare. If a nudge reverses the sign, they could potentially increase social welfare if the reduction in externalities outweighs the loss in consumer surplus. This alternative model generally strengthens the case for taxes. In the cigarette market, taxes only begin to dominate nudges if the deception is substantial. For vaccines, deception makes subsidies consistently outperform nudges. Since nudge effects are too small relative to the significant externality in the energy market, taxes remain superior. Interestingly, highly deceptive nudges can make a policy mix worse than a simple tax because they increase the variance of biases and thus add deadweight loss.

As for the perspective from "Nudges as Moral Taxes", nudges work by imposing direct psychic costs (guilt) or benefits (warm glow) associated with consumption rather than just changing choices through information or bias correction. In the cigarette market, since smoking involves overconsumption, moral taxes impose guilt. This direct welfare loss from reduced consumption likely dominates any welfare gain. Nudges are unattractive and potentially reduced compared to doing nothing. Since vaccination involves underconsumption, moral subsidies provide a warm glow in the vaccine market. Nudges inducing a warm glow become more competitive with financial subsidies, potentially outperforming them if the warm glow effect is significant. In the electricity market, like cigarettes, electricity involves overconsumption, so moral taxes also impose guilt. Interpreting nudges as a moral tax model makes them less attractive in the cigarette and energy markets while potentially improving their stance in the vaccine market.

Research Study: Adverse Selection and Switching Costs in Health Insurance Market: When Nudging Hurts

Handel (2011) also found empirical evidence that "nudging" didn't always improve market efficiency and social welfare by investigating the health insurance market. The researcher analyzed a natural experiment with intricate economic models.

Adverse Selection and Poor Consumer Choices

In the beginning, the author introduced two leading impediments to an efficient insurance market. The first economic issue is the problem of adverse selection, which is a well-studied economics topic. This problem resulted from the nature of asymptotic information in the insurance market. Since the insurers often have insufficient information about the consumers' health conditions and risks, they cannot perfectly price the risk characteristics. This will lead to the result that healthier people tend to choose less coverage since they believe they don't need comprehensive coverage and rather save more money, while the more vulnerable individuals have no choice but to enroll in the comprehensive plans considering future risk. In turn, the premium price of more comprehensive plans goes up, potentially causing consumers to choose less coverage that they prefer.

The other problem is less studied, but equally important, is poor consumer choice. A lot of studies are summarized by Thaler & Sunstain (2008). Insurance plans are often complicated in their contexts, which make it challenging to evaluate the true costs and benefits between different plans. As the Affordable Care Act (ACA) has been proposed, policy makers aimed to improve consumer's choice by emphasizing and clear standardized benefit descriptions. If consumers lack the information or ability to adequately choose their insurance plan, this will lead to immediate inefficiency and long-term losses.Thereby, the core of the research is to empirically investigate how switching costs, one of the barrier of choice inadequacy, interact with adverse selection in the context of an employer-sponsored insurance setting. The cost incurred in switching between insurance plans not only includes monetary cost, but also covers the transaction cost, which includes the time and effort spent in finding the optimal insurance plans. Researching new plans and comparing them with old plans are time consuming and burdensome. Thus, people often end up sticking to the old ones even though more cheaper options are available. This inertia will also lead to market inefficiencies.

These phenomena have been studied separately, but their interaction has been neglected . The author stresses the importance of researching interaction, since choice adequacy affects plan enrollment, which in turn impacts average costs and premiums. Therefore, policies designed to improve consumer choice by reducing switching costs might have an ambiguous welfare effect. Better decision-making conditional on fixed prices could be offset by increased adverse selection if prices adjust. This contradicts previous work where improving choices is typically seen as positive for welfare.

Empirical evidence

To study the potential of nudge, the researchers used data that contained a natural experiment to identify switch costs from consumer preferences. A firm implemented a major change in the offerings of health insurance plans, forcing the employees out of the health plans they had been enrolled in and required them to actively choose a new health plan. The firm also made substantial effort to ensure that employees made active choices at t0 by continuously contacting them via physical mail and email to communicate both information about the new insurance program and that there would be no default option. In the subsequent years, the insurance plans remained the same and the employees could just continue on their plans without taking any actions.

In the experiment, the company changed the menu by offering 3 PPO plans (They also offered 2 HMOs but respondents who chose either one were dropped by the author since these are covered by different network) that provided different coverage in deductible and out of pocket maximums but have the same network of doctors and hospitals. (PPO options after the menu change by their respective individual-level deductibles: PPO250, PPO500 and PPO1200. PPO1200 is paired with a health savings account (HSA) option that allows consumers to deposit tax-free dollars to be used later to pay medical expenditures.) After the deductibles are paid, PPO250 has a coinsurance rate of 10% while the other two plans have rates of 20%, implying they have double the marginal price of post. Both PPO250 and PPO500 have the same at-fee copayment structures for pharmaceuticals and physicians on visits, while in PPO1200 these apply to the deductible and coinsurance. Healthy employees should have chosen PPO500 and sick employees PPO250.

The results are compared between the year before the reform, the “active year” that employees are required to choose their plan, and the later years when they can passively stay in their plans. The data is robust as it included detailed information not only about the plan choice, but also medical spendings for each employee and their family. This makes it possible for the researcher to understand if the plan choices are rational and economically sensible and allow the researcher to estimate each family’s expected health costs.

Before digging through economic modeling, the paper presented the descriptive tests based on the data alone as a baseline. The data showed the presence of both switching costs and adverse selection without modeling. Three pieces of evidence are found. Firstly, there is evidence that employees once locked in one of the insurance plans won't change thereafter. New employees who joined the company when the reform became active tended to choose plans that were cheaper; old employees who had been with the company often didn't choose these cheaper plans since the inertia kept them from reacting to the new prices. Secondly, despite the fact that the employees have been provided with lower total employee expenditures, they still stick to their old choice. One of the insurance plans, PPO250, became less favorable compared to other plans in the second term(t1). However, the majority of employees stayed in the more expensive plan. Evidence has also shown that people who switched tended to change in other plans, suggesting some people are more proactive. Thirdly, people who had chosen more comprehensive plans generally have much higher medical bills, and they consistently chose more comprehensive plans after the reform. This indicates a pattern of how consumers make decisions: they initially select plans based on their health risk, then stick with the plan they had chosen. This suggests evidence of high switching costs, and this reduces the extent of adverse selection if people have to make an active choice in t1.

To go even further, the paper built an economic model to quantify the effects of how families choose health insurance. The author develops a structural choice model with three main components: switching costs, family-level risk preferences, and individual-level ex-ante cost projections. The complicated model puts emphasis on several characteristics of decision-making in order to quantify welfare to the largest extent. Firstly, The choice model assumes families choose a plan to maximize their expected utility, conditional on their risk tastes and health risk distributions. In essence, families pick the plans which give them the highest "expected utility" as a dominant criteria. They choose the best protection against financial risk from medical bills and the price of the plan jointly. Secondly, families are assumed to have Constant Absolute Risk Aversion (CARA) preferences. People dislike uncertainty about big medical bills more than saving a bit on premiums if they stay healthy. Thirdly, as a strong assumption, the model assumes families know their future health expenditure risk distribution and have full knowledge of plan characteristics. Switching costs are modeled as an incremental cost which occurs when switching, aligning with prior literature. Lastly, the model also assumed that consumers do not forecast structural or strategic change, myopic decision-making, due to the difficulty consumers would face in such forecasting. The cost model assumes no moral hazard and no family-specific private information beyond what's in the detailed claims data.

The estimates reveal large switching costs. This substantial average switching cost suggested that a plan would have to be better in expected value for the average person to bother switching from their default option. Families with dependents had higher switching costs than single employees. Interestingly, employees who used FSAs, which require active enrollment each year, had lower switching costs. This indicates they are more engaged consumers.

Evidence also shows moderate risk aversion. People value insurance for protecting against large financial shocks from illness. Also, there was a strong underlying dislike for the PPO 1200 plan, which was a high-deductible plan linked to a Health Savings Account (HSA). Even after accounting for an employer contribution to the HSA, people seemed to avoid this plan more than its financial characteristics alone would suggest. This could be due to the complexity or hassle cost of managing a HSA. The robustness of these findings was checked against different modeling assumptions. The main results about high switching costs generally held up.

The paper then analyzes welfare to study the impact of policy intervention where consumer switching costs are reduced as two scenarios.

1. Scenario 1: No Plan Re-Pricing(The "Naive" View): Here we assume that insurance prices stay fixed even when people switch plans. Switching costs are reduced and people can adjust more readily to price change. In this sense, people can make better choices for themselves and move to plans that are a better fit for their risk and the (fixed) prices. Thus, overall welfare goes up.
2. Scenario 2: Endogenous Plan Re-Pricing (Primary Policy Analysis): In reality, insurance premiums for a given plan are based on the average cost of the people enrolled in that plan in the previous year. The firm also used a subsidy structure to make sure the employees faced the full marginal cost difference between plans. In this case, healthier people are freer to move. They are able to quickly switch from more comprehensive plans like PPO250 to less comprehensive, cheaper plans like PPO500. This leaves the more comprehensive plans (PPO250) with an even sicker, higher-cost group of enrollees than before. Consequently, the next year's premium payment for PPO 250 has to go up significantly to cover these higher average costs. PPO500 might become even cheaper if it attracts a healthier pool. This wedge between PPO250 and PPO500 will be widened. This price gap between comprehensive and basic plans leads to even more adverse selection in subsequent years. The more comprehensive plan, PPO 250, can enter a "death spiral", where it becomes too expensive to afford, and only be selected by the very sick individuals.

While individuals who switch might benefit from the reduction of switching costs, many who didn’t switch are made worse-off. The overall economic welfare actually decreases substantially.. Reducing switching costs by 75% led to an average welfare loss of about 7.7% of premiums paid. This policy effectively doubled the existing welfare loss caused by adverse selection in the baseline scenario. This negative outcome generally holds even if we assume that reducing the "effort" of switching is a benefit to consumers. The harm from worsened adverse selection outweighs the benefits of lower switching costs, unless the switching costs are entirely eliminated and the full value of that elimination is counted as a welfare benefit.

The paper discusses various economic reasons why switching costs exist, beyond just monetary fees. These include (1) transaction/Search Costs: The time and mental effort to learn about complex plan details, compare them, and fill out forms. (2) Learning Costs: Effort to understand how to use a new plan's features. (3) Rational Inattention/Fixed Re-optimization Costs: consumers pay some transaction cost to re-optimize and change plans after acquiring information. (4) Psychological Biases/Inertia: A simple preference for the status quo or brand loyalty. .

The finding that "nudging" can hurt doesn't mean we should keep consumers in the dark. It means that policymakers need to keep in mind policies that improve choices (like the ACA's push for clearer information or employer decision-support tools) have the potential to worsen adverse selection if we overlook the pricing mechanisms. Mechanisms like risk adjustment become even more important in a world with lower switching costs to prevent these adverse selection. The central economic insight is that in markets with imperfect information and the potential for adverse selection, interventions that seem beneficial can have unintended negative consequences when prices are allowed to adjust. The paper calls for a more formal analysis of the interactions between policies to improve decision-making and other key market regulations in settings with potential adverse selection.

Policy Implication: Expanding from Nudge

As nudging has been the center of debate in behavioral economics, public policy has expanded beyond the initial concept of “nudge”.

“Nudge plus” is a modified version of classic nudge that incorporates the elements of reflection(Banerjee & John, 2021). This approach considers the individual’s self-awareness and internal deliberation to make it more effective and legitimate. This intends to extend the effect of public policy and generate long-term, persistent, and sustainable behavior change. Nudge plus builds on recent work combining heuristics (Type 1 processing) and deliberation (Type 2 processing). Nudge Plus can be delivered as a one-part device, where the nudge and reflection are intrinsically combined, or as a two-part device, where a nudge is extrinsically combined with a deliberative instrument prompting reflection. Examples include a dual self-pledge device with commitments catering to both short-term and long-term preferences, or a GPS device with an AI assistant. Unlike a classic nudge that relies on automatic responses, nudge plus a stress trigger of reflection. For instance, a commitment device becomes a nudge plus when it also provides information about its aims or offers feedback. The "plus" element aims to enhance autonomy by making individuals conscious of the nudge and allowing them to "own the behavior change process".

"Budge" policies are also a category of behavioural economic interventions. Unlike nudges, which primarily target internalities and the demand side of economics, budge focuses on imposing regulation against harms imposed by the supply side, particularly to counter negative externalities. A budge involves "openly regulating against these activities, or requiring organisations to use behavioural economic-informed interventions that are expected to be beneficial to their clientele" (Oliver, 2015). For example, if confectionary companies exploit present bias by paying substantial money to make supermarkets place their goods near the cashier, a budge policy could regulate this. Budges are explicitly regulatory and are concerned with situations where private organizations might harmfully manipulate consumers.

"Boost" policies stem from the simple heuristics program and aim to extend individuals' decision-making competencies (Grüne-Yanoff & Hertwig, 2016). In contrast to nudge, which exploits cognitive biases, boosts seek to improve skills, knowledge, and decision tools, or restructure the environment so existing skills can be more effectively applied. The goal is to empower people to make better decisions by fostering competencies of decision making in general (statistical literacy) or a professional (medical) context. Boosts assume that decision-makers have competencies that can be improved and are not merely "somewhat mindless, passive decision makers". While nudges often work by influencing System 1 thinking, boosts focus on enhancing the individual's adaptive toolbox of heuristics and making them smarter in specific situations. In comparison, nudge plus and boost direct decision-making by enhancing decision-maker’s knowledge and consciousness, while budge resort to the regulatory approach from the supply side.

Conclusion

Behavioral economics enriches the toolbox in public policy to correct psychological biases by devising choice architecture and directing individuals towards better decisions. This approach, rooted in libertarian paternalism, aims to improve welfare while preserving freedom of choice However, the effectiveness is not always positive. This review demonstrates that nudging, while robust in theoretical reasonings, may have unforeseen consequences. The critical analyses of List et al. (2023) and Handel (2011) serve as powerful reminders that interventions have potential to harm the economy.

Handel's work illustrates how a nudge can exacerbate adverse selection and decrease overall welfare if not carefully integrated with market pricing mechanisms, Similarly, List et al. highlight that the relative effectiveness of nudges versus traditional instruments like taxes depends on the specific context, particularly the balance between internalities and externalities, and the nature of bias heterogeneity. The "targeting ratio" they propose offers a valuable framework for such assessment.

The evaluation the paper uses underscores net welfare effects. It is straightforward, however, within an intricate system of market dynamics and individual differences, the criteria may not always hold. This complexity brings a critical question for policymakers: how far should they analyze when considering behavioral tools? The answer is not always clear-cut when we extend the boundary of analysis. This can reveal further unintended consequences or ripple effects, both positive and negative. Moreover, the success or failure of a nudge and its ultimate impact can significantly depend on the chosen metrics and the Key Performance Indicators (KPIs). A policy might achieve a narrow target but comes at the cost of the other beneficial activities, and thus, the overall welfare calculation becomes ambiguous. The very definition of "making choosers better off, as judged by themselves" necessitates a careful consideration of what aspects of well-being are being prioritized and measured.

In response to the challenges and further understanding of behavior bias, the toolkit available to policy has been expanded. Beyond the classic nudge, approaches like "nudge plus" (emphasizing reflection), "budge" (regulating against harmful supply-side practices), and "boost" (fostering decision-making competencies) offer alternative or complementary strategies.

Ultimately, the critical examination of nudge effectiveness leads to the discussion of the scope of deliberate manipulation and ethical boundaries. While nudge can potentially guide individuals toward better outcomes, it should not erode an individual’s autonomy. The objective should always be to simplify choice and foster better understanding. Instead of merely directing individuals, policymakers must achieve these aims by staying true to their roots of libertarian paternalism, ensuring that interventions empower individuals to make their own informed decisions.

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